

Linear Algebra 1. (03/09/25)

$$S_n = \textcircled{1} + 2 + 3 + 4 + \dots + n$$

Problem-1:

$$S_n = \frac{n}{2} [a_0 + a_n] \rightarrow \textcircled{A.P.}$$

$$S_n = \frac{n}{2} (1+n) = \frac{n(n+1)}{2}$$

$$A = \{a_{ij}\}_{n \times n} = \begin{cases} i; i=j \\ 0; i \neq j \end{cases}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 2 & 0 & 0 & \dots & 0 \\ 0 & 0 & 3 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \vdots & \dots & n \end{bmatrix} \begin{matrix} \rightarrow i=1 \\ \rightarrow i=2 \\ \rightarrow i=3 \\ \dots \\ i=n \end{matrix}$$

ex.:

$$[A]_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$S_2 = 1 + 0 + 0 + 2 = 3$$

$$S_n = \frac{n(n+1)}{2} = \frac{2 \times 3}{2} = \textcircled{3} \text{ Ans}$$

Problem-2

$$A = \{a_{ij}\}_{n \times n} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

$$[A]_{n \times n} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{bmatrix}$$

$$S_n = 1 + 1 + 1 \dots + 1$$

$$S_n = \frac{n}{2} (1+1) = \boxed{n}$$

Ex 4 ::

$$A = \{a_{ij}\}_{4 \times 4} = \left\{ \frac{i^2 - j^2}{2} \forall i, j \right\}$$

The sum of all elements of matrix is 0 ✓

$$\sum a_{ij} = 0$$

Symmetric matrix $\Rightarrow A^T = A$
 Skew " " $\Rightarrow -A^T = A$

$$[A]_{4 \times 4} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}_{4 \times 4}$$

$$= \begin{bmatrix} 0 & -\frac{3}{2} & -4 & -\frac{15}{2} \\ \frac{3}{2} & 0 & -\frac{5}{2} & -6 \\ 4 & \frac{5}{2} & 0 & -\frac{7}{2} \\ \frac{15}{2} & 6 & \frac{7}{2} & 0 \end{bmatrix}$$

skew symmetric

Problem-V

$$A = \{a_{ij}\}_{n \times n} \\ = \{k + i + j\}$$

Problem-VI

$$A = \{a_{ij}\}_{n \times n} \\ = \begin{cases} 2 \cdot (3) & i \neq j \\ 0 & i = j \end{cases}$$

Find the sum of all elements for the matrix?

Key points :-

1) Sum of Sequence of n -natural numbers.

$$S = [n(n+1) \cdot (2n+1)]/6 \quad \text{coding}$$

2) Sum of squares of first n even no's is

$$S = [2n(n+1) \cdot (2n+1)]/3 \quad \checkmark$$

3) Sum of squares of first n odd numbers,

$$S = [n(2n+1)(2n-1)]/3 \quad \checkmark$$

Note :-

Sum of n terms of a G.P given by:-

$$S_n = \begin{cases} \frac{a(1-r^n)}{1-r} & \text{if } |r| < 1 \\ \frac{a(r^n-1)}{r-1} & \text{if } |r| > 1 \\ a_n & \text{if } |r| = 1 \checkmark \end{cases}$$

Sum of an infinite G.P is given by

① $S_\infty = \frac{a}{1-r} ; |r| < 1$.

② $S_\infty = \infty, |r| \geq 1$

Trace of a Matrix

The sum of the leading diagonal elements of a Square Matrix A is called a trace of A .

denoted by:- $\text{tr}(A)$.

$$A = (a_{ij})_{n \times n}$$

$$\text{tr}(A_{n \times n}) = \underbrace{a_{11} + a_{22} + \dots + a_{nn}}_{\text{K(A)}} = \sum_{i=1}^n a_{ii}$$

Properties of trace of Matrix

if A & $B \rightarrow$ Square matrix.

① $\text{tr}(\lambda A) = \lambda \text{tr}(A)$ ✓ $\because \lambda$ is a scalar

② $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$,

③ $\text{tr}(A-B) = \text{tr}(A) - \text{tr}(B)$

④ $\text{tr}(AB) = \text{tr}(BA)$ ✓

⑤ $\text{tr}(AB) \neq \text{tr}(A) \cdot \text{tr}(B)$

⑥ $\text{tr}(I_n) = n$.

$$\begin{array}{c} 4 \quad 3 \quad 2 \quad 1 \\ \swarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ 1 \quad 2 \quad 1 \quad 1 \end{array}$$

$$A = \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} \quad ; \quad B = \begin{vmatrix} 4 & 3 \\ 1 & 2 \end{vmatrix} \quad \vee \quad \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix}$$

$$AB = \begin{vmatrix} 11 & 12 \\ 5 & 5 \end{vmatrix}$$

$$BA = \begin{vmatrix} 11 & 15 \\ 5 & 5 \end{vmatrix}$$

$$\Rightarrow \text{tr}(AB) = 11 + 5 = 16$$

$$2) \text{tr}(BA) = 11 + 5 = 16$$

$$\begin{aligned} \text{tr}(A) &= 2 + 1 = 3 \\ \text{tr}(B) &= 4 + 2 = 6 \end{aligned}$$

solⁿ //

$$\therefore \text{tr}(AB) = \text{tr}(BA) \neq \text{tr}(A) + \text{tr}(B)$$

Example:

① A square Matrix is defined as-

$$A = [a_{ij}]_{3 \times 3} = \begin{cases} 0; i < j \\ i^2 - j^2; i \geq j \end{cases}$$

The trace of Matrix is $\frac{12}{0}$

② $A = [a_{ij}]_{3 \times 3} = \begin{cases} i^3 - j^3 \forall i, j \end{cases}$ ✓

→ 0

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\text{tr}(A) = 2 + 4 + 6 \\ = 12$$

Problem:-

(IITJEE-2021)

$$\text{Let, } A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix} \quad \text{--- (i)}$$

$$\text{and } 2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix} \quad \text{--- (ii)}$$

if $\text{Tr}(A)$ denotes the sum of all diagonal elements of $\text{Mat-}A$.
Then $\text{Tr}(A) - \text{Tr}(B)$ has value equal to —

(A) 1 (B) 2
(C) 3 (D) 0

M-1

(i) + 2(ii)

$$\text{Tr}(A) - \text{Tr}(B) = 1 - (-1) = 2$$

$$A + 2B + 2A - 2B = \begin{bmatrix} 3 & 0 & 5 \\ 10 & -5 & 15 \\ -5 & 3 & 3 \end{bmatrix} = 3A$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ -1 & 1 & 1 \end{bmatrix} \rightarrow \text{Tr}(A) = 1$$

$$\text{(ii)} \rightarrow 2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \text{Tr}(B) = -1$$

$$B = 2A - \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow B = \begin{bmatrix} 0 & 2 & -1 \\ 2 & -1 & 0 \\ -2 & 1 & 0 \end{bmatrix}$$

$$\begin{aligned} & \text{tr}(A+2B) = \text{tr}(A) + \text{tr}(2B) \quad \text{--- ①} \\ & \text{tr}(A+2B) = 1 - 3 + 1 = -1 \\ & \text{tr}(2A-B) = 2 - 1 + 3 = 3 \\ & \text{tr}(2A) - \text{tr}(B) \quad \text{--- ②} \end{aligned}$$

$$\begin{aligned} \text{tr}(A) + 2\text{tr}(B) &= -1 \quad \leftarrow \\ 2\text{tr}(A) - \text{tr}(B) &= 3 \\ 4\text{tr}(A) - 2\text{tr}(B) &= 6 \\ 5\text{tr}(A) &= 5 \\ \text{tr}(A) &= 1.0 \end{aligned}$$

$$\begin{aligned} & 1 + 2\text{tr}(B) = -1 \\ & 5\text{tr}(B) = -1 \\ & \text{tr}(B) = -1/5 \\ & \text{tr}(A) - \text{tr}(B) = 3 \\ & \text{tr}(A) - (-1/5) = 3 \\ & \text{tr}(A) = 3 - 1/5 = 2.8 \quad \checkmark \end{aligned}$$

Next class $\rightarrow$ Matrix operations
(Addition/Subtraction/
Multiplication) etc

